Qn. Set Code-7(SOT)

*Semester: 5th*
*Programme: B.Tech*
*Branch: Computer Engg.*

# AUTUMN END SEMESTER EXAMINATION-2025

**5th Semester B.Tech**

## HIGH PERFORMANCE COMPUTING

## CS30005 / CS 3010

**(For 2024 & Previous Admitted Batches)**

Time: 2 Hours 30 Minutes | Full Marks: 50

***Answer any FIVE questions.***
***Question paper consists of two SECTIONS i.e. A and B.***
***Section A is compulsory.***
***Attempt any Four question from Sections B.***
*The figures in the margin indicate full marks.*
*Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.*

## SECTION-A

1. Answer the following questions. [1 × 10]

(a) Consider an instruction pipeline with 5 stages that take 7 nsec, 4 nsc, 3 nsec, 8 nsec, and 5 nsec respectively. The delay of an inter-stage register stage of the pipeline is 2 nsec. What is the approximate speedup of the pipeline in the steady state under ideal conditions as compared to the corresponding non-pipelined implementation?

(b) Write down the corresponding assembly language MIPS instruction for the following machine code given in the hexadecimal form:

**018E7822**

**016D6020**

The opcode of load, store, add and sub is 35, 43, 32 and 34 in integers.

(c) Consider two processors P1 and P2 executing the same instruction set. Assume that under identical conditions, for the same input, a program running on P2 takes 25% less time but incurs 30% more CPI (clock cycles per instruction) as compared to the program running on P1.If the clock frequency of P1 is 2 GHz, then calculate the clock frequency of P2 (in GHz)?

(d) What hardware implementation exists in MIPS to avoid structural hazard while referencing memory and register?

(e) What are the two ways of handling cache read miss operation?

(f) Calculate the percentage of miss prediction accuracy using 2-bit dynamic branch prediction scheme for the following outcome of a branch:

T, NT, T, T, T, NT, NT, NT, T, T, NT, T, NT, T, NT, NT

Assume the initial prediction is **strongly NOT taken**.

(g) In which stage of the scoreboard approach, the output dependency and anti-dependency have been resolved?

(h) Distinguish between super-scalar and super-pipeline processor.

(i) In a real time system, it's observed that a CPU generate 200 memory references out of which 10 misses in L1 cache, 6 misses in L2 cache. The miss penalty for L2 cache to memory is 200 clock cycles. The hit time for L2 cache and L1 cache are 5 clock cycles and 1 clock cycle respectively. What is the local miss rate for L2 cache and AMAT?

(j) Explain Compulsory, Capacity and Conflict misses in Cache.

## SECTION-B

2. (a) Derive the overall speed up gained by Amdahl's law. [5]

Suppose a processor spends 62% of its time in memory operations (both I- cache and D-cache access together) when running a given program A and its manufacturers make a change that improves its performance on memory access by a factor of 6. The program originally took 212 sec to execute. What is the speed up from the old system to the new system? What fraction of its execution time does the new system spend in memory operations?

(b) Find out the total no of clock cycles required to execute the following instructions without and with operand forwarding and draw the **time and space diagram for both the cases**? [5]

LD R11, 50(R2)

DADDIU R11, R11, #10

SD R11, 40(R2)

DADDIU R5, R2, #4

DSUB R4,R3,R5

3. (a) Explain how reservation stations and Common Data Bus(CDB) used in the Tomasulo's approach are useful in resolving the WAW and WAR hazard with the following example for each hazard.[Assumption: [5]

Initially all the registers are available, i.e., not waiting for any functional units. The number of functional units is also sufficient enough to accommodate all the instructions given.]

Find out the total number of clock pulse required to execute the following instructions using Tomasulo's approach.

ADD Fl, F2, F3

STORE Fl, 5(R4)

ADD F1, F5, F6

SUB F7, F1, F8

STORE Fl, 6(R5)

ADD F1, F9, F10

(b) Explain loop unrolling and perform loop unrolling for the following code fragment [5]

represented in high level language-

```
for ( i=0; i<=100; i++)
{
    a[i] = a[i] - 4;
}
```

This code subtracts a scalar to a vector. Here each element in the array is of type double, occupying 8 bytes. In the pipeline branch condition is checked in the Execution stage. Ignore the delays introduced by the branch instruction.

Assume the following latency in the example:

| **Instruction producing the result** | **Instruction using the result** | **Latency in clock cycles** |
|---|---|---|
| FP ALU op | Another FP ALU op | 3 |
| FP ALU op | Store double | 2 |
| Load Double | FP ALU op | 1 |
| Load Double | Store double | 0 |
| Integer operation | Integer op | 0 |

4. (a) For a given machine, following are the different type of instructions and their corresponding clock cycles:- [5]

| | |
|---|---|
| LOAD | 5 |
| STORE | 4 |
| ALU | 4 |
| BRANCH | 3 |
| JUMP | 3 |

If a program has 50% ALU instructions, 15% LOAD instructions, 25% STORE instructions, 8% BRABCH instructions and 2% JUMP instructions, then evaluate the CPI overall. If the same program has 2000 instructions and the speed of the computer is 2.4GHz, calculate the time taken to execute the program.

(b) How a centralized shared memory system is different from distributed shared memory system. Explain with a suitable diagram. [5]

5. (a) Assume a fully associative write back cache with many cache entries that starts empty and block size is 4 Bytes and word length is 1 byte. Below is a sequence of memory operations (the address is in square brackets). [5]

Write Mem [2500]

Read Mem [2400 ]

Write Mem [3000]

Read Mem [2000 ]

Read Mem [6000 ]

Write Mem [3000]

Write Mem [2000]

Read Mem [3000]

Write Mem [3000]

What are the numbers of hits and misses (inclusive read and write) when using no-write allocate and write allocate?

(b) What are the various optimizations techniques applied for reducing the hit time, cache miss penalty & Reduce Miss Rate? [5]

6. (a) Assume two cache memory specifications. A split cache which consists of 16KB instruction cache and 16KB data cache and 32KB unified cache. The cache memory hit time is 2 clock cycles, miss penalty is of 85 clock cycles. [5]

For the above cache memory specifications a simulator showed the following conditions like:

I) 15 miss per 1000 instructions for data cache.

II) 30 miss per 1000 instructions for instruction cache.

III) 45 miss per 1000 instructions for the unified cache.

If there are 35% Load/Store instruction then calculate the miss rate and average memory access time (AMAT) for split and unified cache memory.

(b) Explain the working and implementation of snoopy based protocol with state diagram in details. [5]

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